Date: May 15, 2026
To: Prof Nate Sniadecki, Vinh Nguyen, Zachary Lewandowski
From: Kit Nguyen
Subject: Analysis of Rigging Plate

EXECUTIVE SUMMARY

The rigging plate was analyzed to determine if it is safe to use. This was done through ANSYS FEA analysis of the plate with 3 loads of 12 kN each acting on the loading rings. To ensure that the max von Mises stress was accurate in the analysis, different mesh sizes spanning 3 orders of magnitude were used to test for convergence. The results were that under these conditions, the rigging plate is safe to use as the max von Mises stress was 237 MPa while the Yield Strength of the aluminum used for the plate was 500 MPa, meaning that factor of safety was 2.1.

INTRODUCTION

The rigging plate used for industrial lifting, firefighting, and rescue is shown below in **Figure 1**.



Figure 1: Rigging plate

The purpose of this analysis was to determine the stress concentrations that appear when pressure is applied to the plate at the 3 locations highlighted in **Figure 2**. Once the concentrations are determined, it can be compared to the yield strength of the material to determine if the material is able to withstand the conditions applied to it.

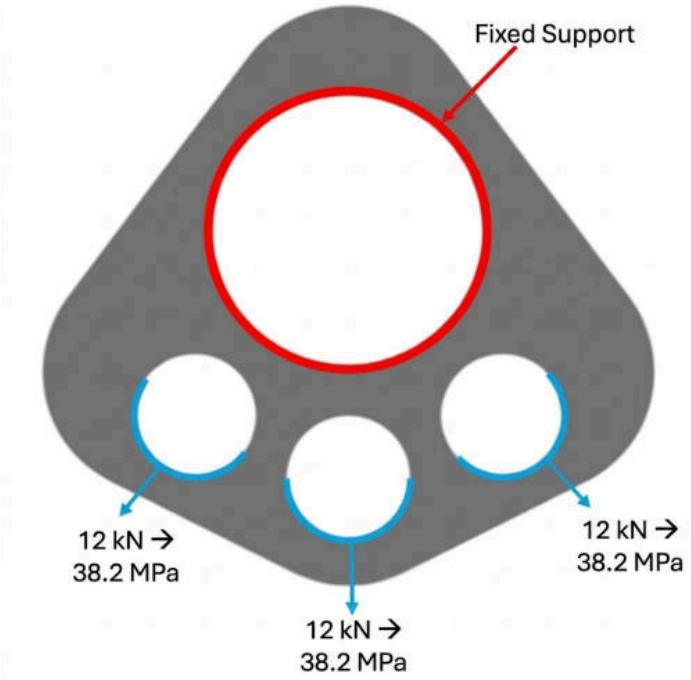


Figure 2: Load locations on Rigging Plate

METHODS

The 3D model of the rigging plate was modeled in ANSYS using the Ansys 2024 R2 Design Modeler. Starting with sketching it in 2D (Sketch and dimensions in **Appendix 1)** and then creating a surface from the sketch with a thickness of 1cm allowed the model to be analyzed in 2D. Once the skit was complete, symmetry was applied through the center axis of the rigging plate (shown in **Figure 3)** since the left and right sides of the plate were identical in geometry and forces/pressures applied. Applying symmetry made the analysis less demanding and simplified the model. The following material properties were used for the aluminum plate **(Table 1)**:

Table 1: Properties of the aluminum

Young's Modulus (GPa)	72
Poisson's Ratio	0.33
Yield Strength (GPa)	500

The model was meshed 10 different times to account for convergence in the stress (meshes in **Appendix 2**). 5 of the meshes were applied globally with the range of meshes targeting at least 3 orders of magnitude. Then, another 5 meshes were applied, but this time at local areas where the stress was concentrated to get a more refined measurement with less computational power.

This one, again, targeted 3 orders of magnitude. The mesh sizes and number of elements are shown in **Table 3 and 4**.

Table 3: Global Mesh sizes and number of elements

Global Mesh	
Mesh Size (cm)	Number of Elements
2	15
1	35
0.2	598
0.1	2299
0.04	13537

Table 4: Local Mesh sizes and number of elements

Local Mesh	
Local Size (cm)	Number of Elements
0.1	457
0.04	1728
0.02	5659
0.01	20216
0.008	31063

As shown in **Figure 2**, the pressure that was applied was 38.2 MPa that was calculated from the force that was applied over the area of the edge, which was from the thickness * circumference of the ½ circle.

$$P = F / A$$
$$F = 12\ kN$$
$$A = C * t = \pi\ cm^2$$
$$P = 38.2\ MPa$$

RESULTS

GLOBAL MESH

Table 5 shows the results of the meshing analysis on a global level (changing the size of all elements).

Table 5: Global Mesh Data

Mesh Size (cm)	Number of Elements	Total Time (s)	Max Von Mises Stress (Pa)
2	15	9	1.00E+08
1	35	9	1.77E+08
0.2	598	9	1.78E+08
0.1	2299	8	2.02E+08
0.04	13537	36	2.23E+08
0.02	53574	163	2.43E+08

Plotting the number of elements vs max von Mises helps determine if the mesh converges or goes to infinity. Looking at just the global mesh plot **(Figure 3)**, it can be seen that the mesh begins to converge between 2299 elements and 13537 elements. The clear indicator for this is that as the mesh elements increase and become more refined, the max von Mises stress begins to level out. Here, we see that by around 5000 elements, increasing the number of elements even more (by 1 order of magnitude) does not have a significant change in getting a more accurate stress reading as it levels out at about 250 MPa.

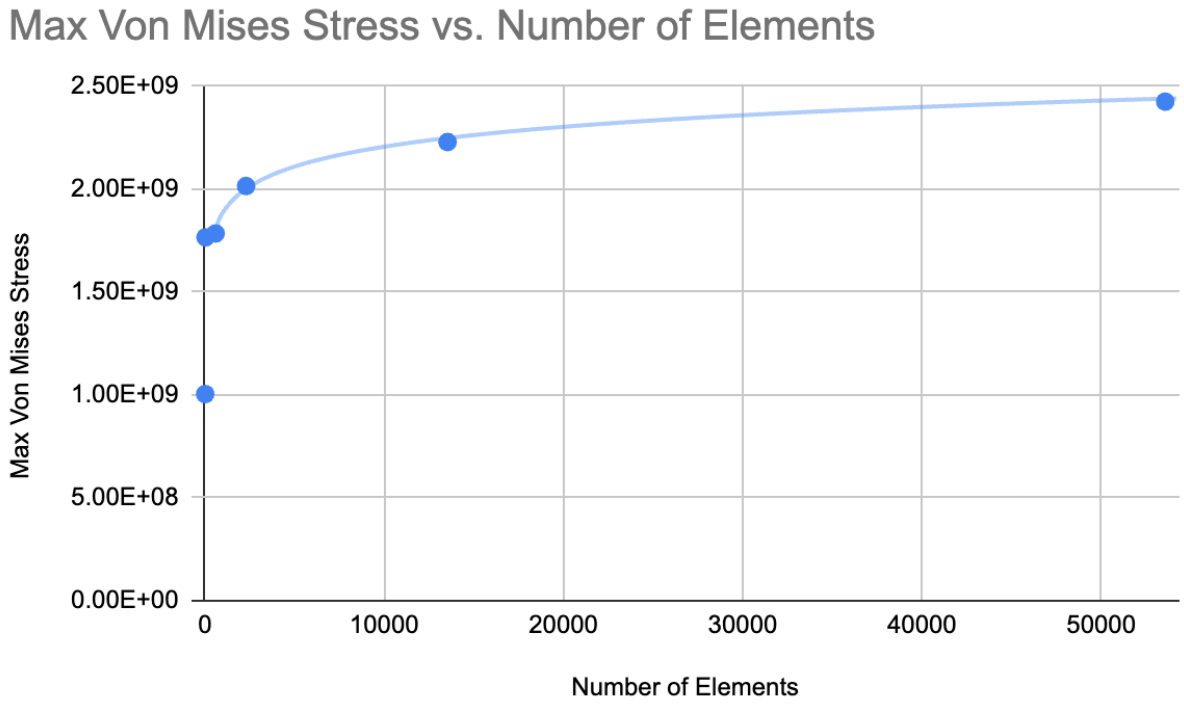


Figure 3: Global Mesh Max Von Mises Stress vs. Number of Elements

A similar plot can be made for the total run time (mesh generation and analysis) versus the number of elements **(Figure 4)**. With an increased number of elements, the time it takes to

generate is exponentially increasing (not converging). Using these two graphs, an optimal mesh size of 10000 elements (around 0.04 cm element size) can be an optimal mesh size for future analysis since it balances the computational power required (therefore total time to generate the mesh) and is beyond where the convergence happens, so we can be sure that the mesh is fine enough to have accurate analysis.

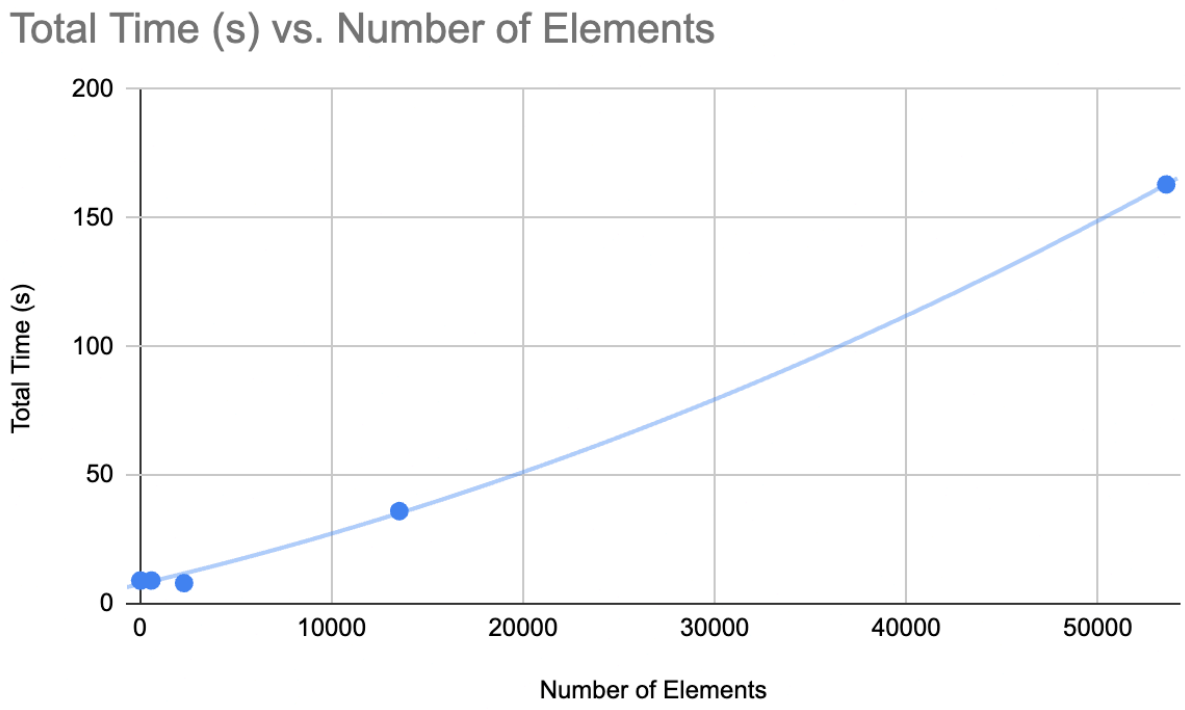


Figure 4: Global Mesh Total Run Time vs. Number of Elements

The locations of max total deformation **(Figure 5)** and maximum von Mises stress **(Figure 6)** is shown below for the optimal global mesh size. The maximum deformation happens at the location that attachments are connected to the rigging plate, where it is 50.452 μm. This makes sense because these locations are closest to where the force is applied so the pulling that force is doing across the entire plate is cumulating to the location that it is being pulled, making it the largest.

Similarly, the maximum von Mises stress occurs on side edges of the holes for attachment because stress is flowing toward the hole, causing it to be concentrated there. Here, the max stress is 222.9 MPa.

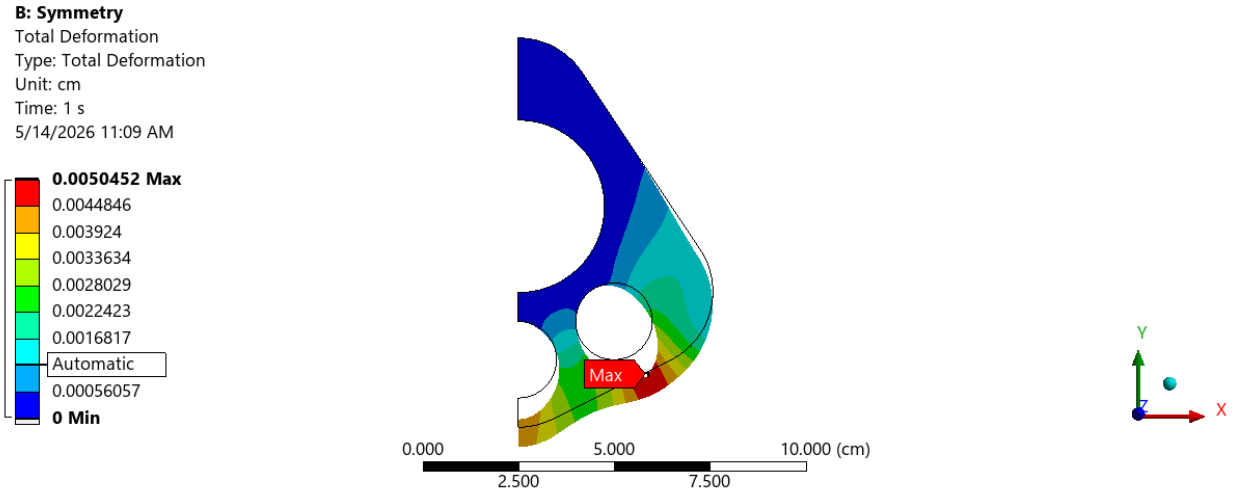


Figure 5: Total deformation for 0.04 cm global mesh

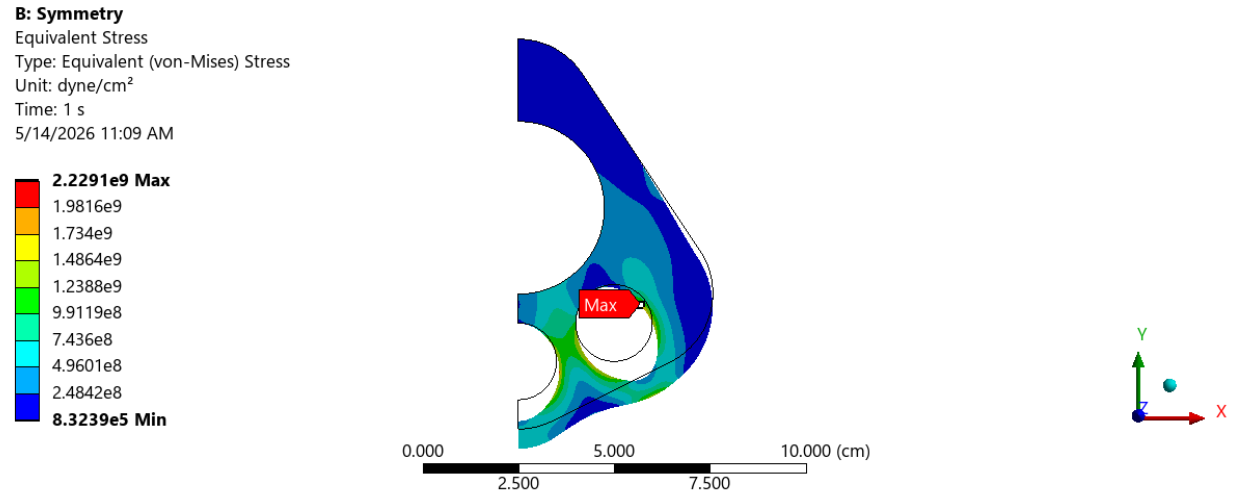


Figure 6: Von Mises stress for 0.04 cm global mesh

LOCAL MESH

In order to get more insight on the max stress concentrations and mesh convergence, a local mesh was used to get more fine elements. This local mesh has a sphere of influence that's 1 cm large and has a global mesh of 0.5 cm. The vertex of the sphere of influence was placed on one of the edges of the attachment area as seen below in **Figure 7.** This location was picked because this is where the maximum von Mises stress was located when the element sizing was the smallest, however both sides of the circle have similar stress magnitudes.

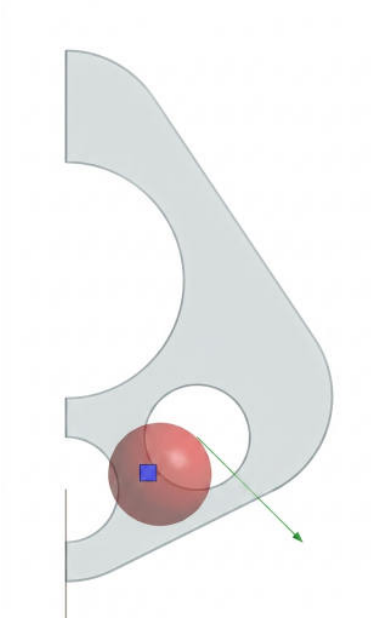


Figure 7: Sphere of influence on the rigging plate

The local mesh analysis resulted in the information in **Table 6** and the graphs in **Figures 8 and 9.**

Table 6: Local Mesh Data

Local Size (cm)	Number of Elements	Total Time (s)	Max Von Mises Stress (Pa)
0.1	457	13	1.99E+08
0.04	1728	9	2.21E+08
0.02	5659	19	2.37E+08
0.01	20216	68	2.56E+08
0.008	31063	110	2.64E+08

The plot of the local max von Mises stress versus number of elements **(Figure 8)** yields similar results to what we had in the global mesh. Convergence begins at around 1728 elements and you can tell that it is converging because the graph is starting to flatten out. The convergence begins to level out at 250 MPa. The local meshing allows for a closer look at what the stress is like at finer element sizes but without being as computationally demanding as a global mesh, so convergence is easier to detect in this case.

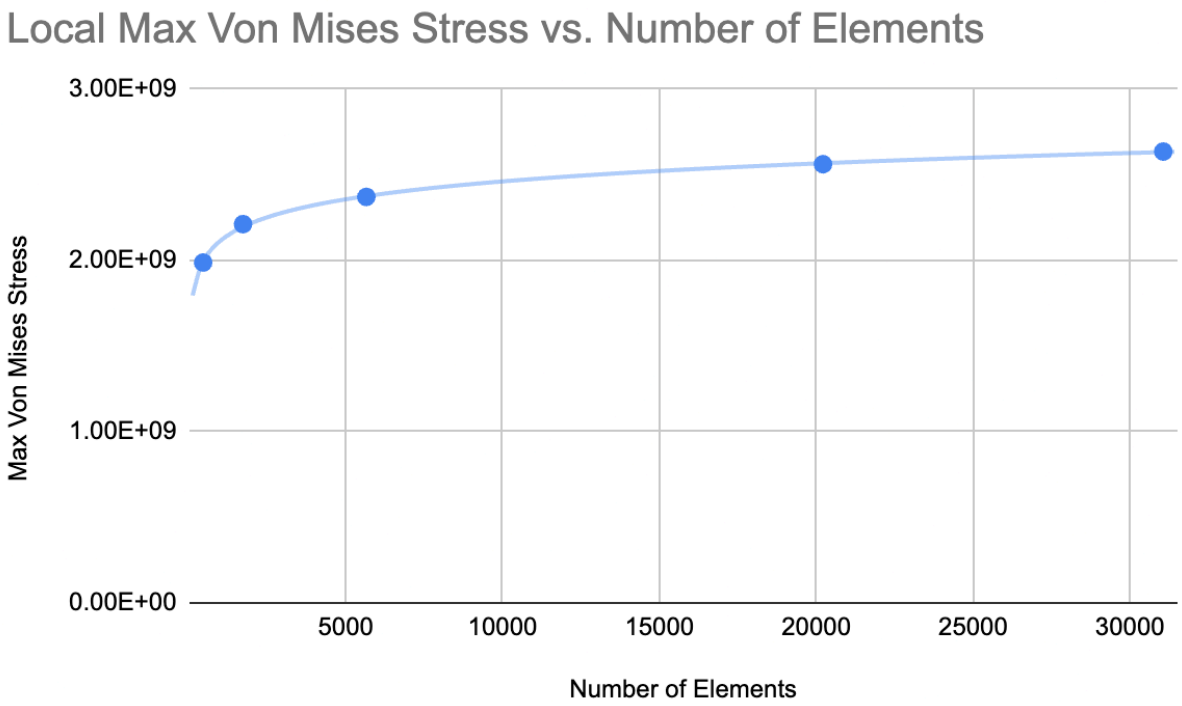


Figure 8: Local Mesh Max Von Mises Stress vs. Number of Elements

Additionally, the local meshing can be analyzed for the total time it takes for the mesh generation and analysis versus the number of elements. When graphed **(Figure 9)**, it can be seen that there is an exponential relationship (meaning it does not converge) between the number of elements and the time it takes, similar to that for the global mesh. Comparing these two graphs, an ideal mesh size would be a local mesh size of 0.02 cm with a 0.5 cm global mesh, resulting in 5659 elements. This mesh sizing would be ideal for minimum computational power while also being after the stress converges, so it would be a fine enough element size to get an accurate reading for the stress.

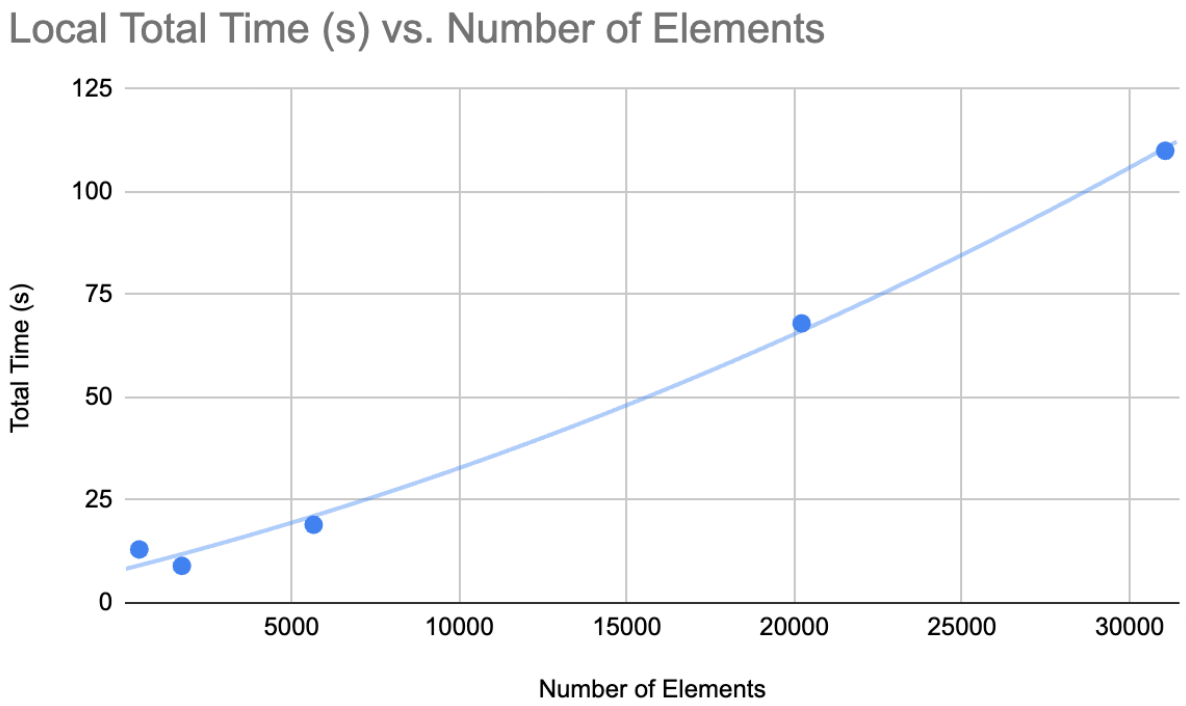


Figure 9: Local Mesh Total Run Time vs. Number of Elements

Total deformation and von Mises stresses are shown below in **Figure 10 and Figure 11** for the optimal local meshing size**.** Mimicking the global mesh, this local mesh has the maximum deformation at the edge of the inner circle on the rigging plate where the load is applied. The value for this is 50.382 μm. This makes sense still because these locations are closest to where the force is applied so the pulling that force is doing across the entire plate is cumulating to the location that it is being pulled, making it the largest. The value doesn't vary as much from global mesh.

For von Mises, the stress is still concentrated at the same spot on the inner edge of the sides of the circle where the load is applied. This makes sense because this is where the tension accumulates in the rigging plate. The max von Mises stress is 237 MPa, which is a slight increase from the global meshing state. This makes sense because the mesh is finer in the local area, so there are more elements concentrated at that spot. With the increase in the elements, the value of the stress gets closer to the convergence value.

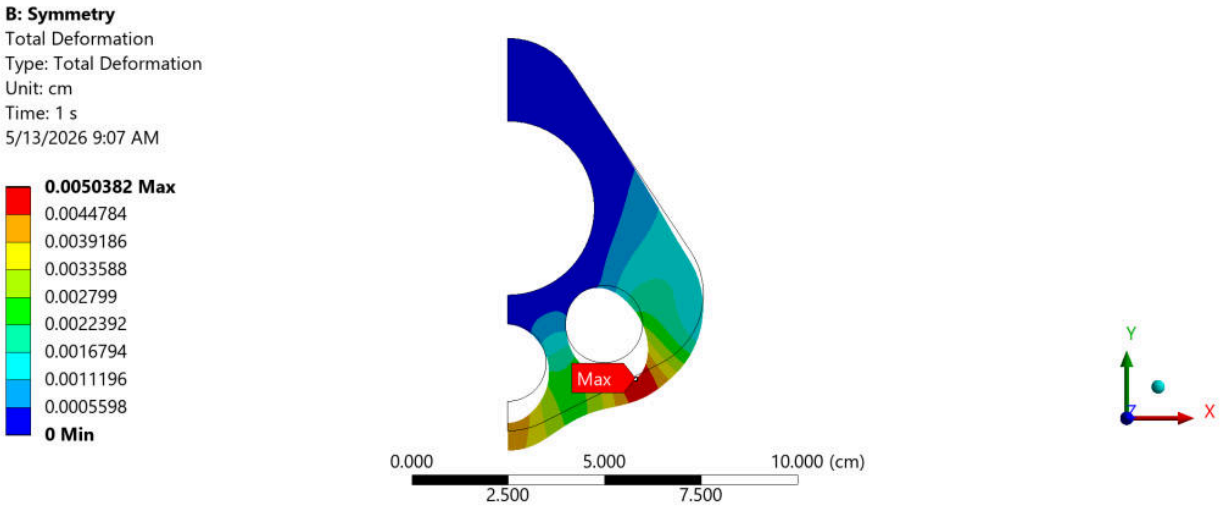


Figure 10: Total deformation for 0.02 cm local mesh

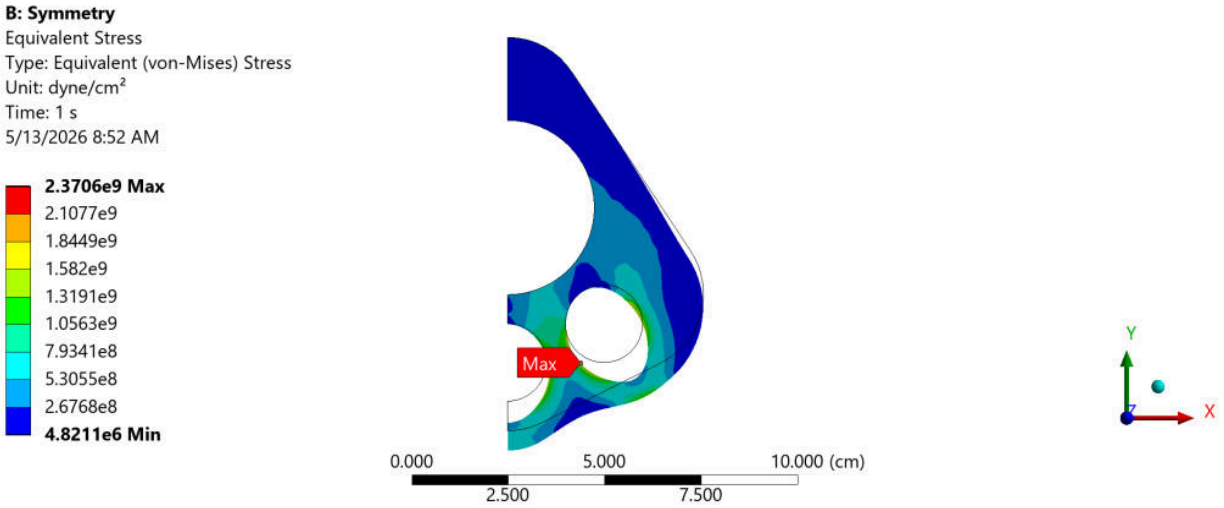


Figure 11: Von Mises stress for 0.02 cm local mesh

SINGULAR FORCE

Another configuration of the forces is shown in **Figure 12**. This configuration is meant to test the stress state when the rigging plate only has 1 load applied to the center of the rigging plate.

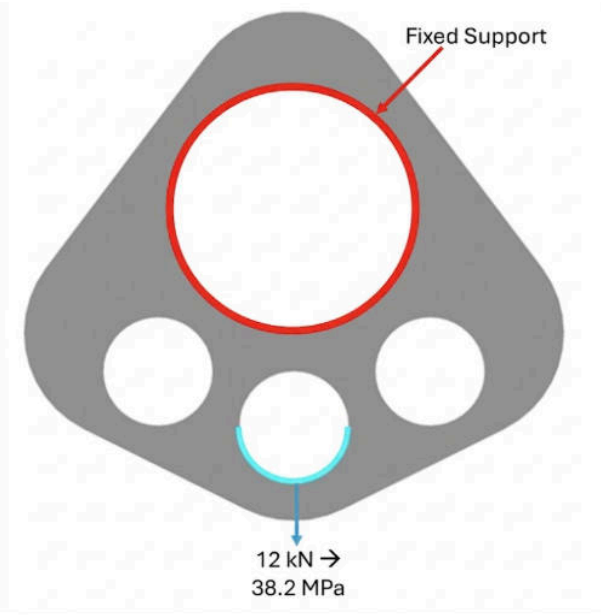


Figure 12: Singular load configuration

In this analysis **(shown in Figure 13)**, the stress concentration is in the side ring of the middle loading area and has a maximum von Mises stress of 331 MPa. The location of the stress makes sense because this is where the tension accumulates in the plate, but one thing that is interesting about this stress state is that it has a higher stress than when 3 loads are applied. A possible reason for this could be that when there are 3 loads, the areas with high stress concentrations are experiencing a little bit of compressions from the other areas, so it cancels out with the tension that its own force is applying to it.

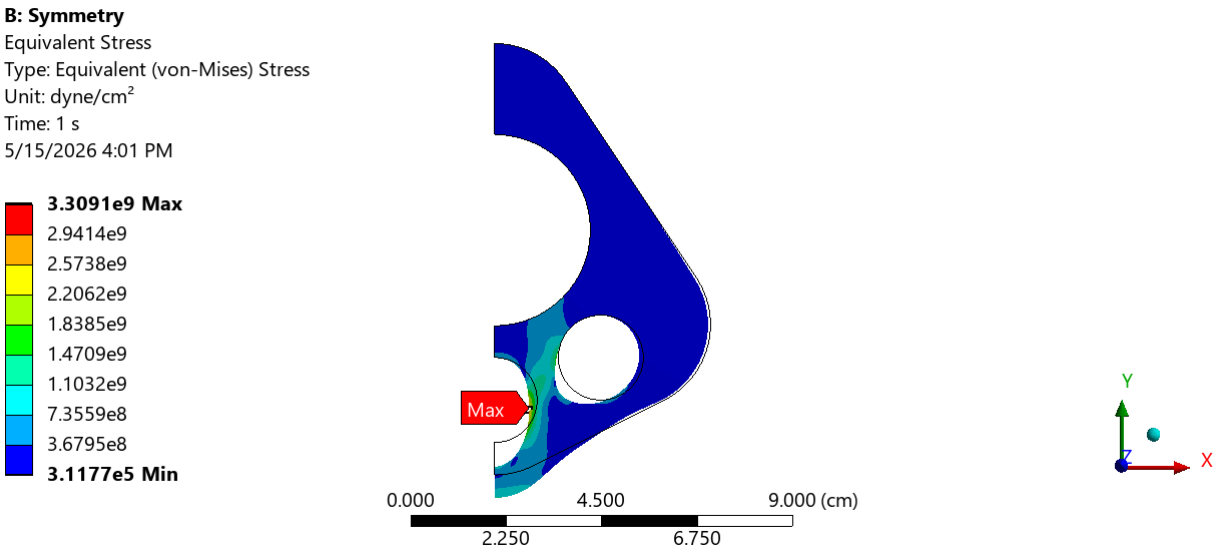


Figure 13: Von Mises Stress of singular load

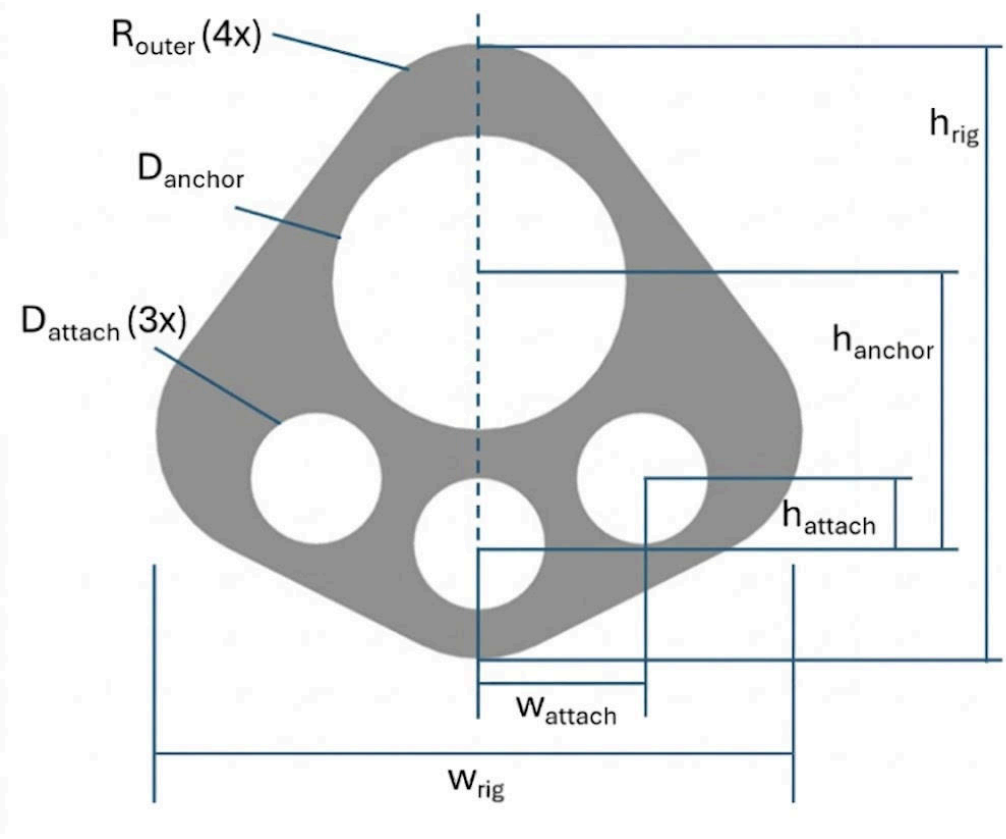
CONCLUSION

After analyzing the von Mises stresses, it is apparent that the rigging plate will not fail under these conditions. For the aluminum material that the rigging plate is made of, the Yield Strength is $\sigma_y = 500\,MPa$. This is double the maximum von Mises stress in from the analysis, which was $\sigma = 237\,MPa$ ($\sigma_y > \sigma$). With a factor of safety of 2.1, it is safe for rigging plate to be used.

In reality, the aluminum should be able to hold up these loads if there was no interesting geometry. Although 38.2 MPa is a lot of pressure, it is distributed along the entire edge of the material. Also, the aluminum is rated for a high yield strength of 500 MPa. The stress concentration is just very large, reaching up to 250 GPa which is almost an order of magnitude larger than the actual load applied. This is because it was amplified by the geometry of the rigging plate, which the load pulls and the pressure is distributed throughout the entire plate. Even in areas where the stress is not at the maximum (i.e. in the area around the ring that the load is applied to), stress concentrations reach up to 80 MPa. Even though the rigging plate has a factor of safety of 2.1, it is still recommended that further precautions are taken, for example making the plate thicker to distribute the force over a larger area.

For the future, further analysis can be done on this by doing analysis on more fine meshes and testing the local meshes in more areas. This will allow us to find the maximum of the convergence and test if the factor of safety dips below unsafe levels.

APPENDIX 1: Geometry and Sketch of Rigging Plate



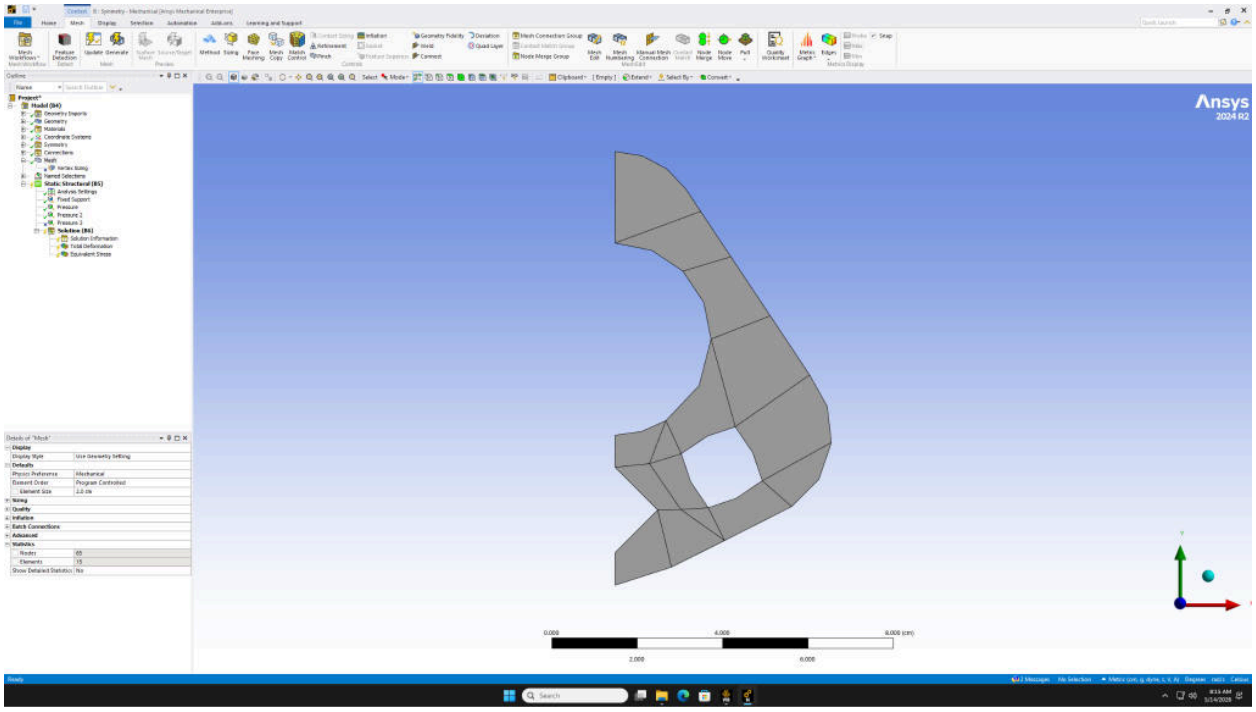
Appendix Figure 1a: Rigging plate geometry

Dimension	Value (cm)
Outer Radius of Rigging Plate, R_{outer}	2
Anchor Diameter, D_{anchor}	4.5
Attachment Diameter, D_{attach}	2
Height of Rigging Plate, h_{rig}	10.16
Width of Rigging Plate, w_{rig}	10.16
Center-to-Center Distance, h_{anchor}	4
Vertical Spacing between Attachments, h_{attach}	1
Horizontal Spacing between Attachments, w_{attach}	2.5

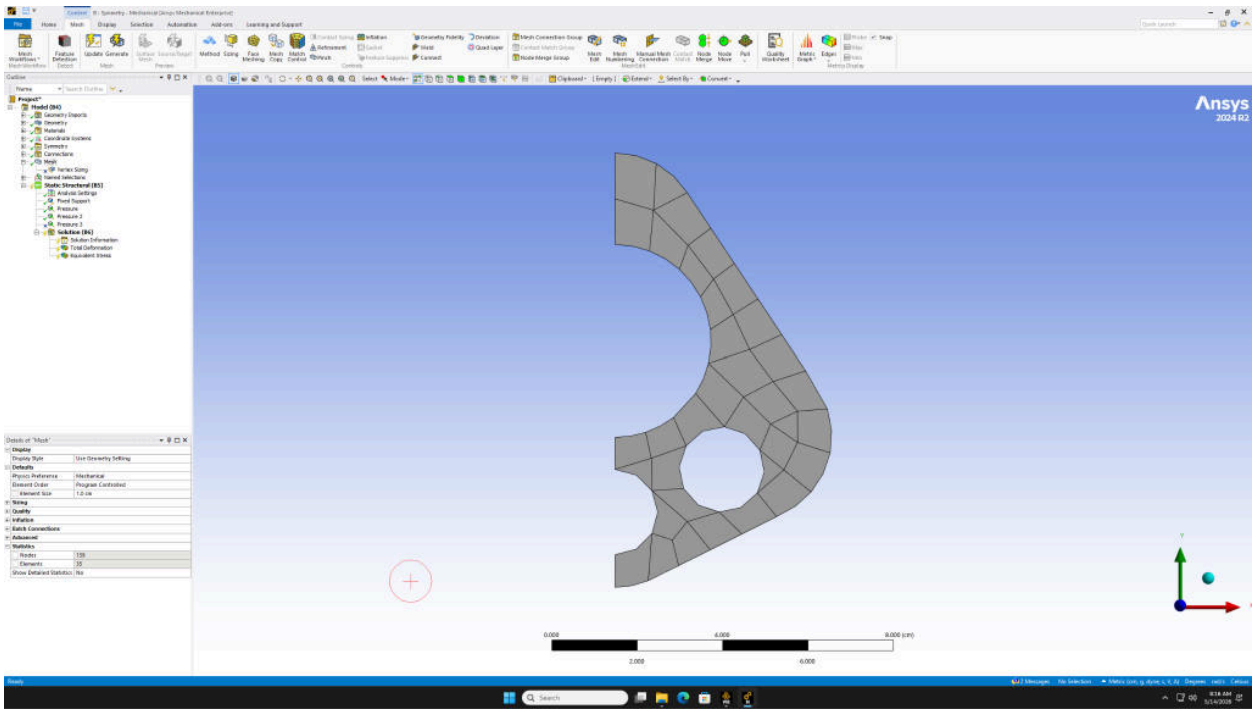
Appendix Figure 1b: Rigging plate dimensions

APPENDIX 2: Meshes Applied to Rigging Plate

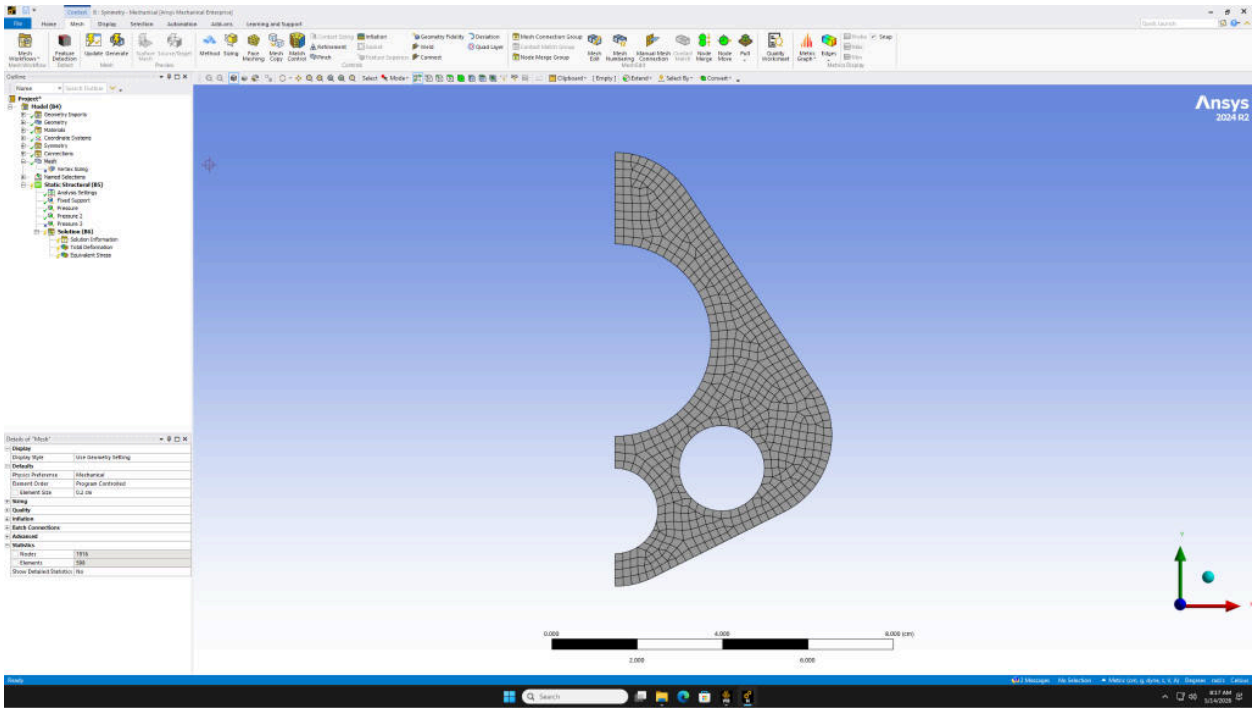
GLOBAL ELEMENTS



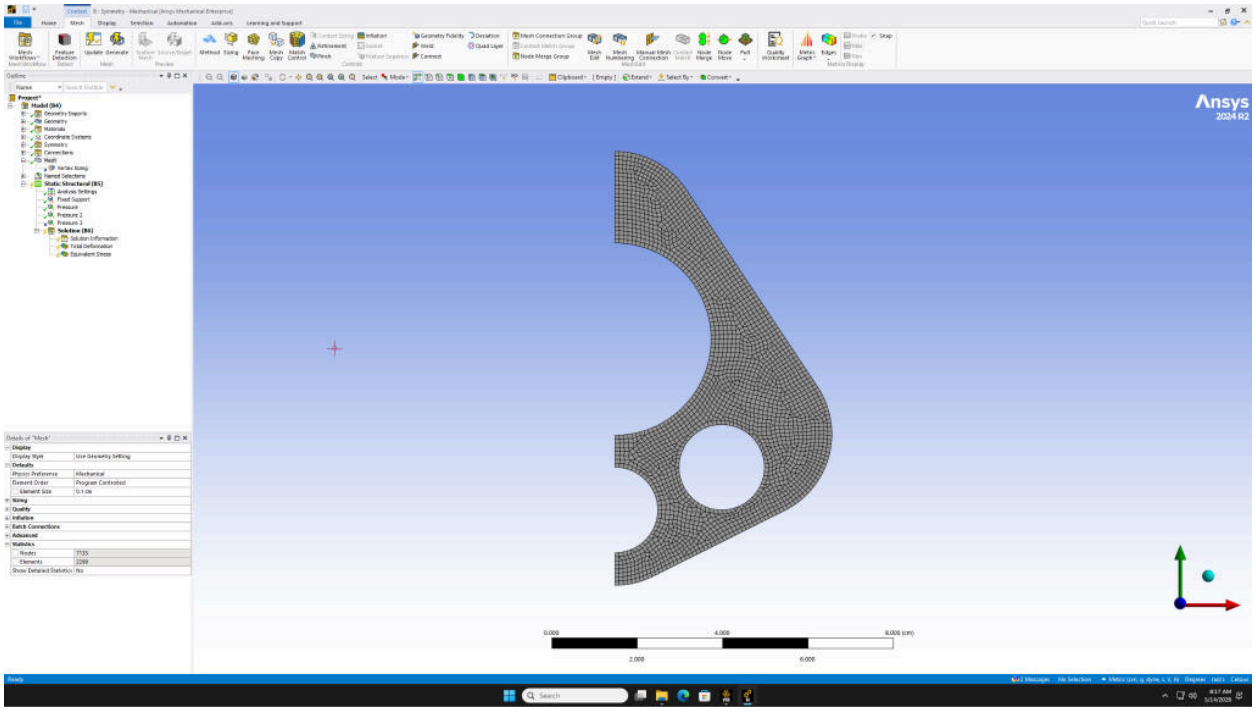
Appendix Figure 2a: 2 cm global mesh



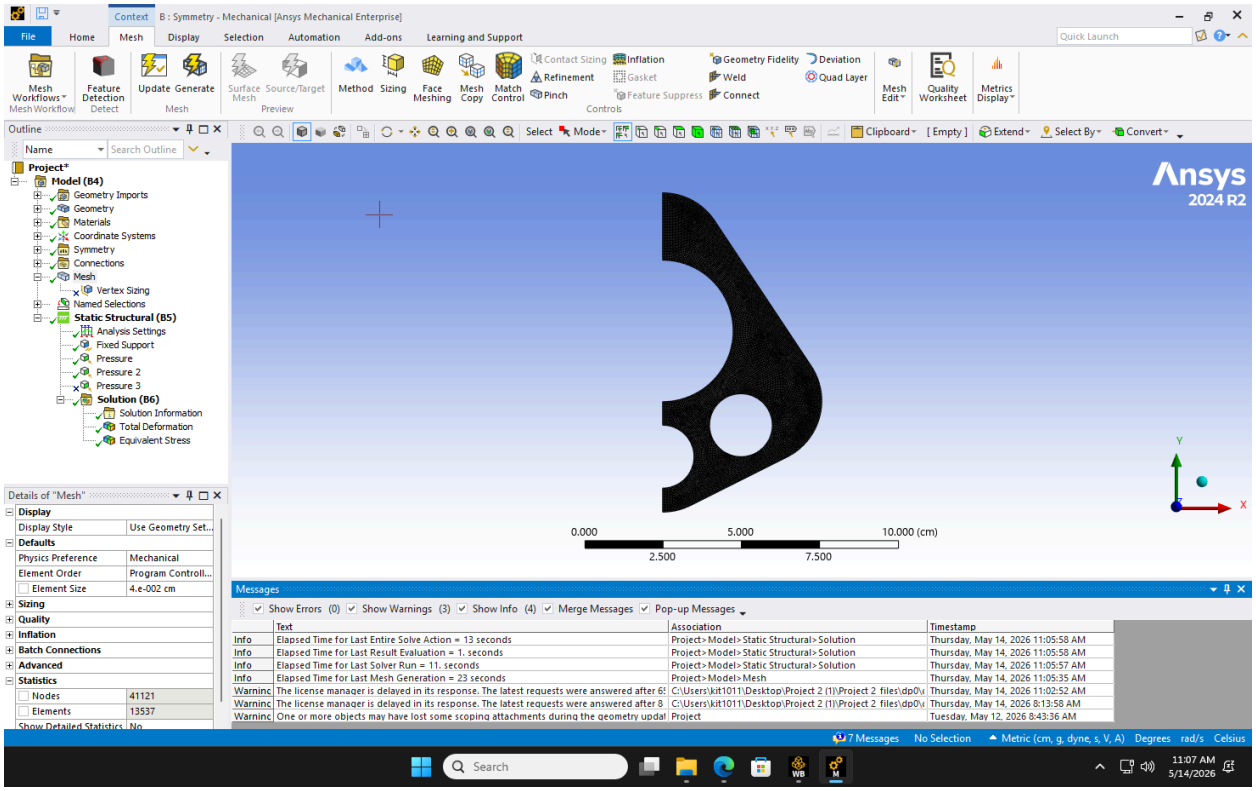
Appendix Figure 2b: 1 cm global mesh



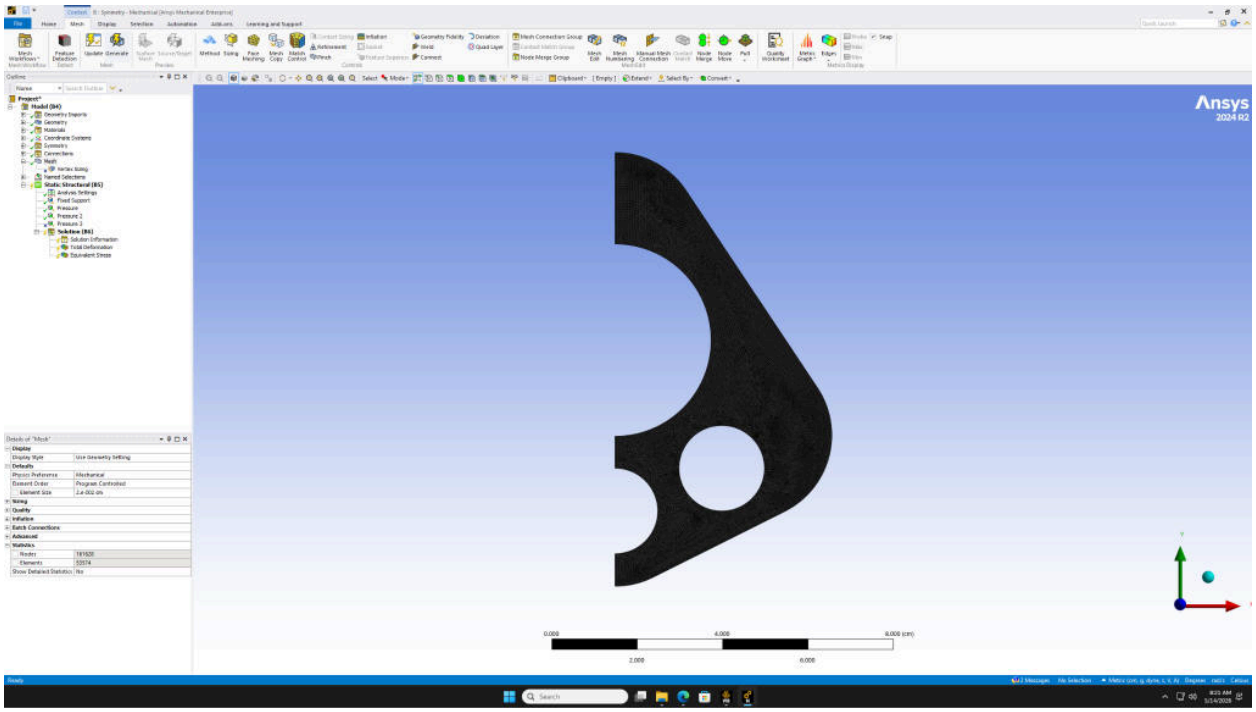
Appendix Figure 2c: 0.2 cm global mesh



Appendix Figure 2d: 0.1 cm global mesh

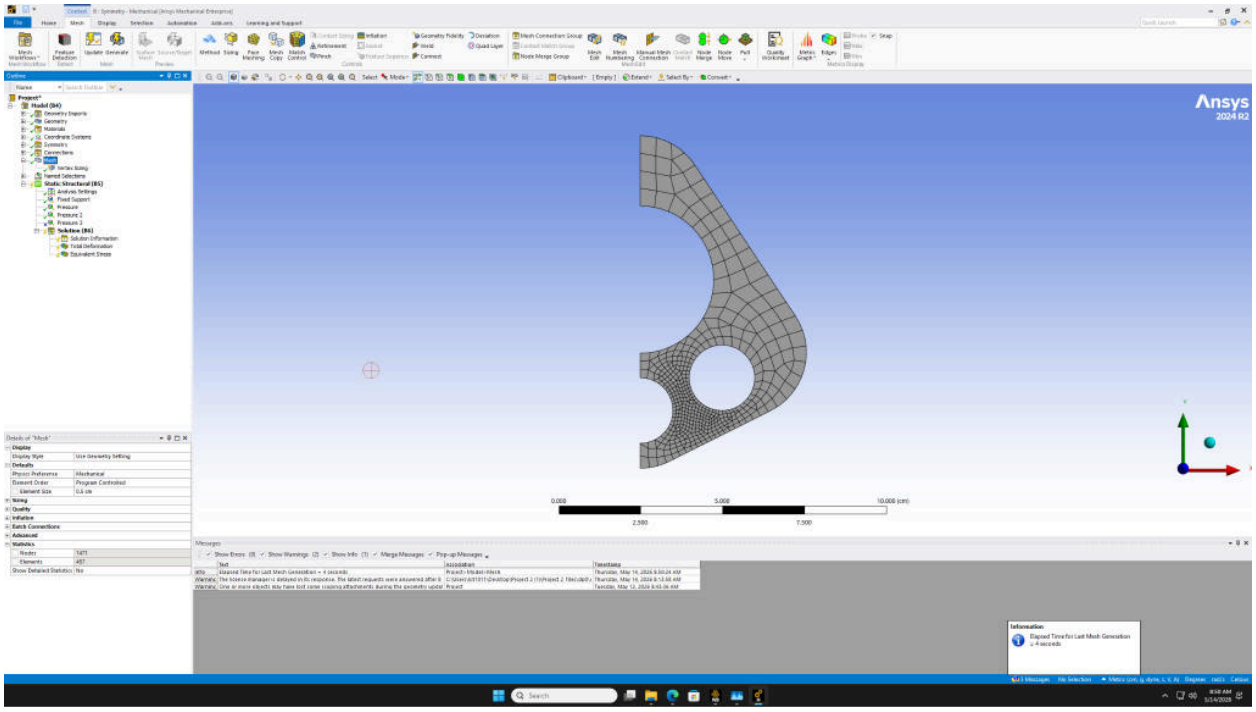


Appendix Figure 2e: 0.04 cm global mesh

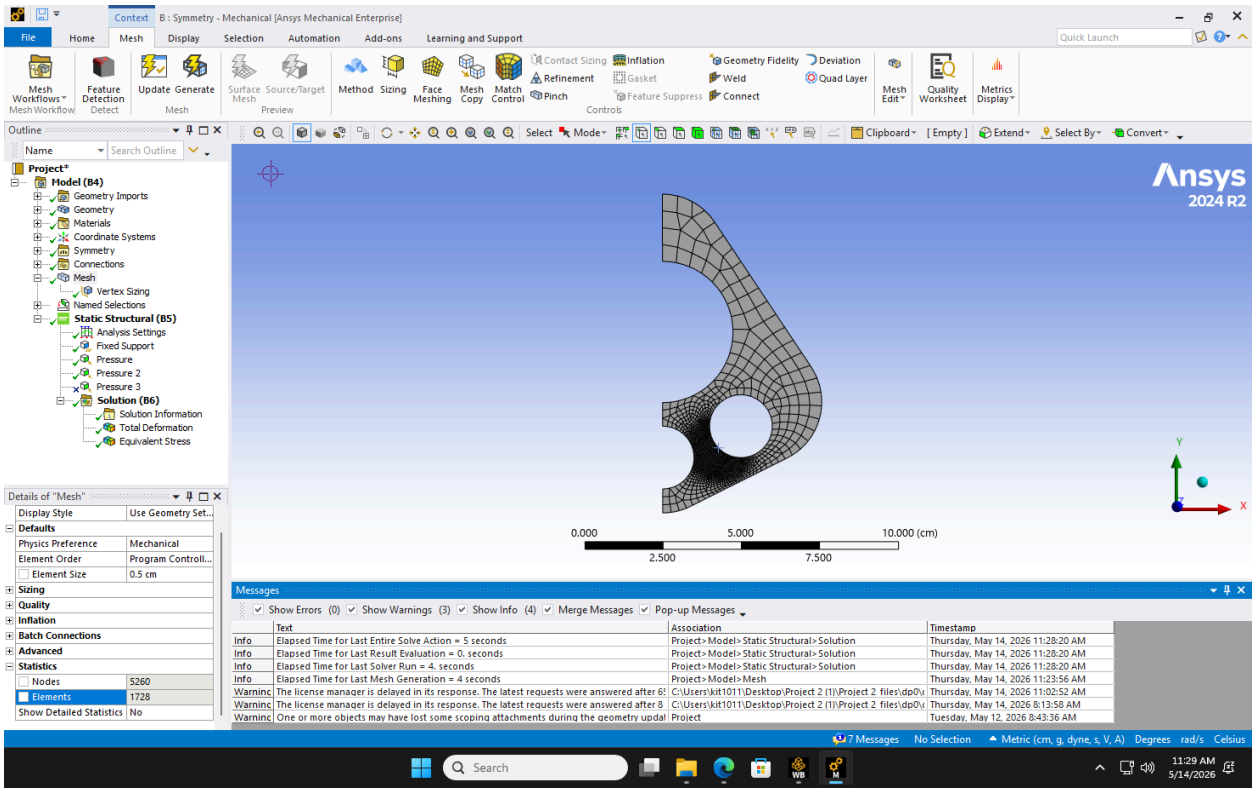


Appendix Figure 2f: 0.02cm global mesh

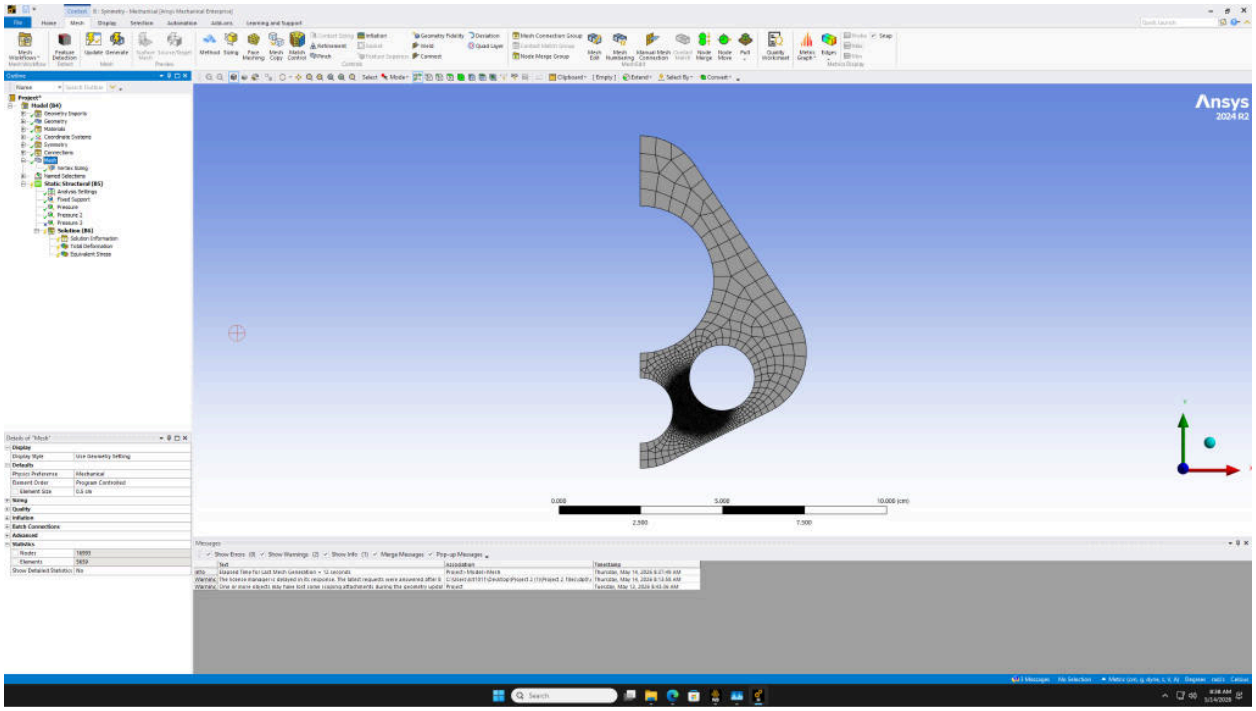
LOCAL ELEMENTS



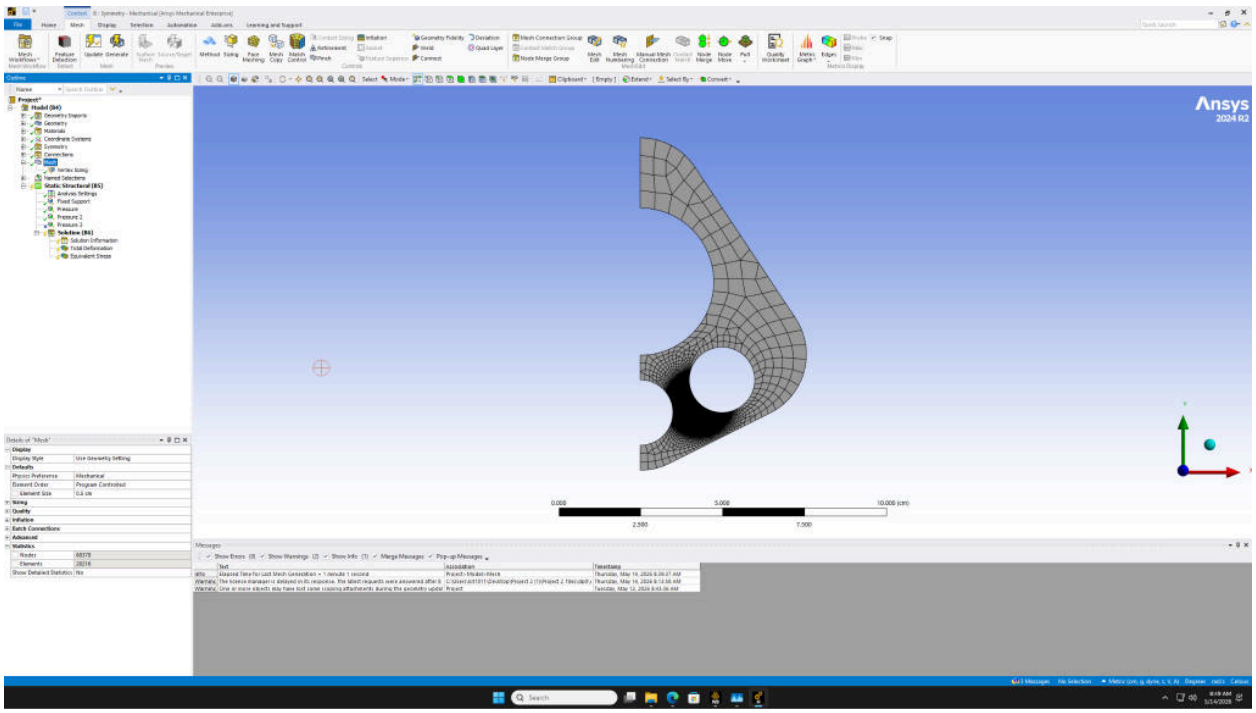
Appendix Figure 2g: 0.1 cm local mesh



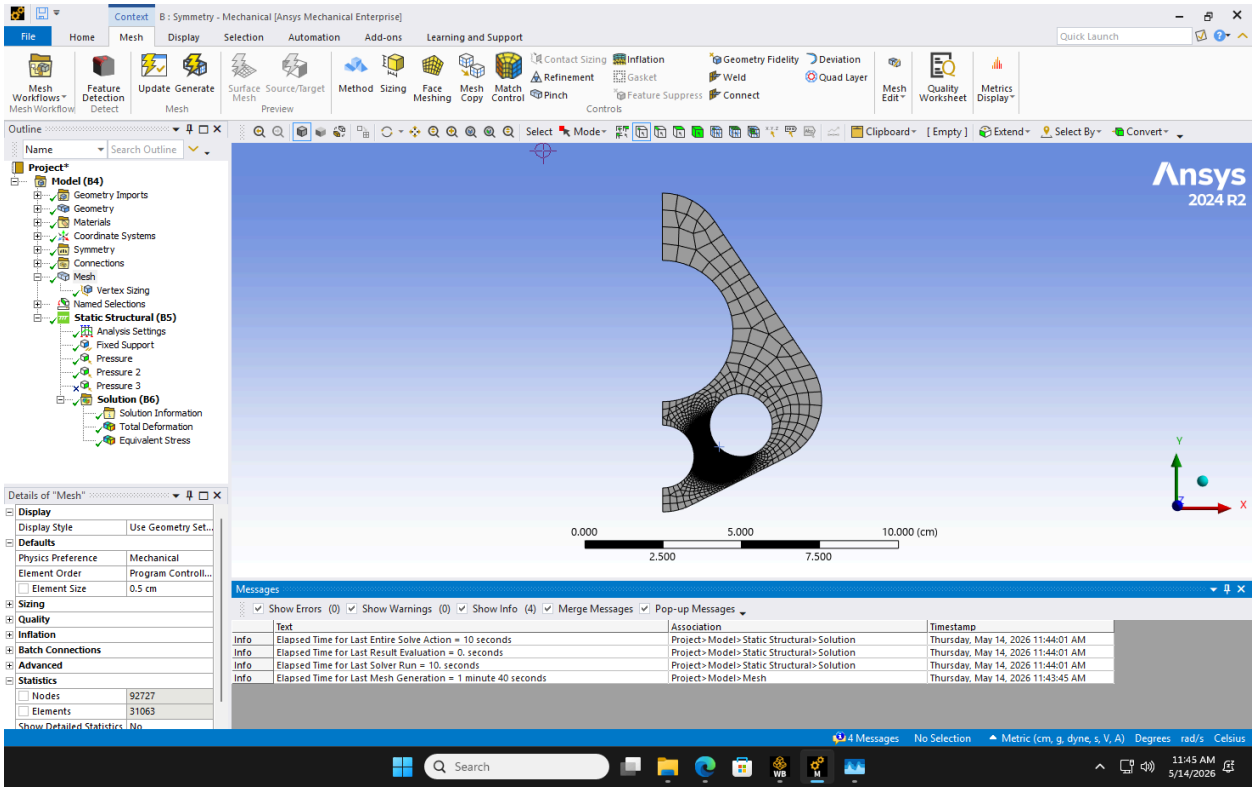
Appendix Figure 2h: 0.04 cm local mesh



Appendix Figure 2i: 0.02 cm local mesh



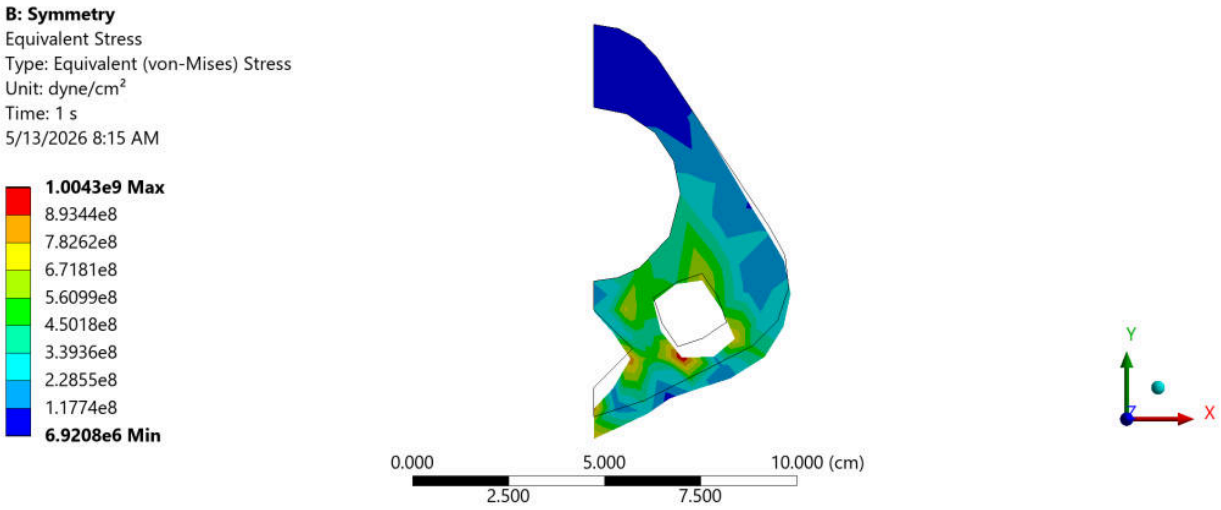
Appendix Figure 2j: 0.01 cm local mesh



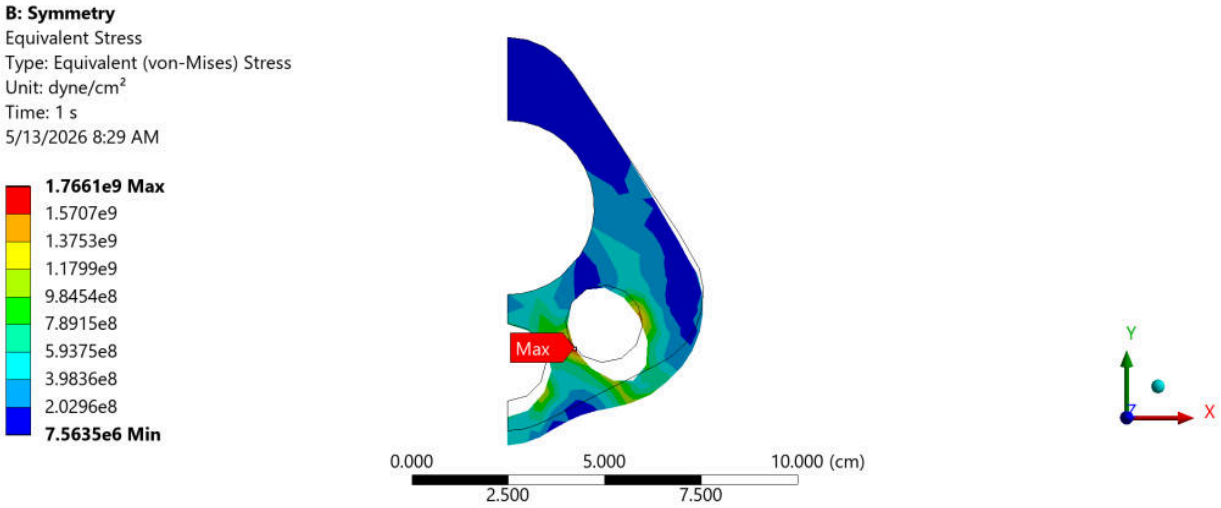
Appendix Figure 2k: 0.008 cm local mesh

APPENDIX 3: Stress Concentrations

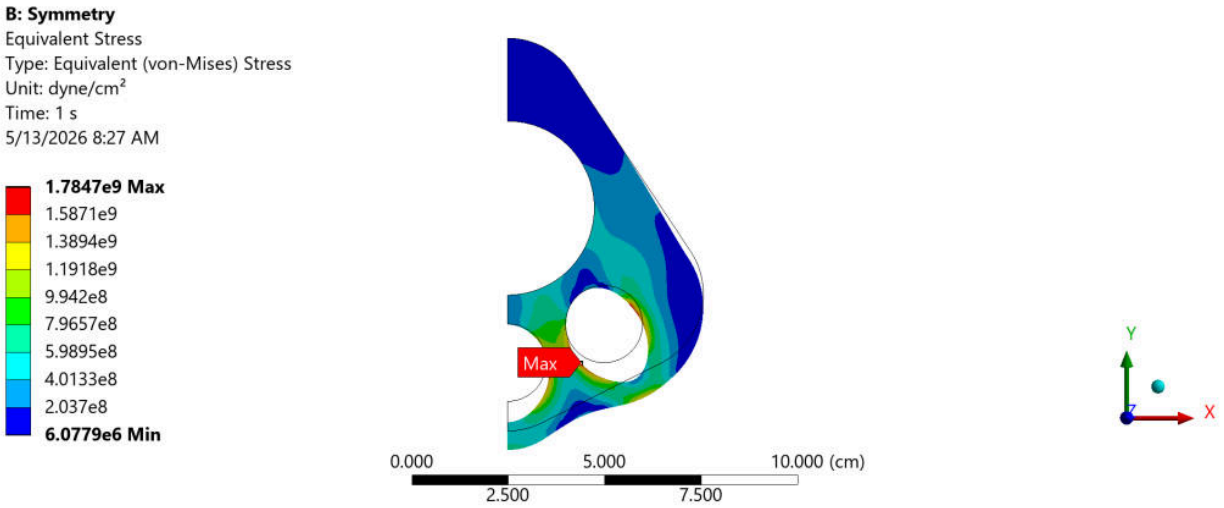
GLOBAL STRESS



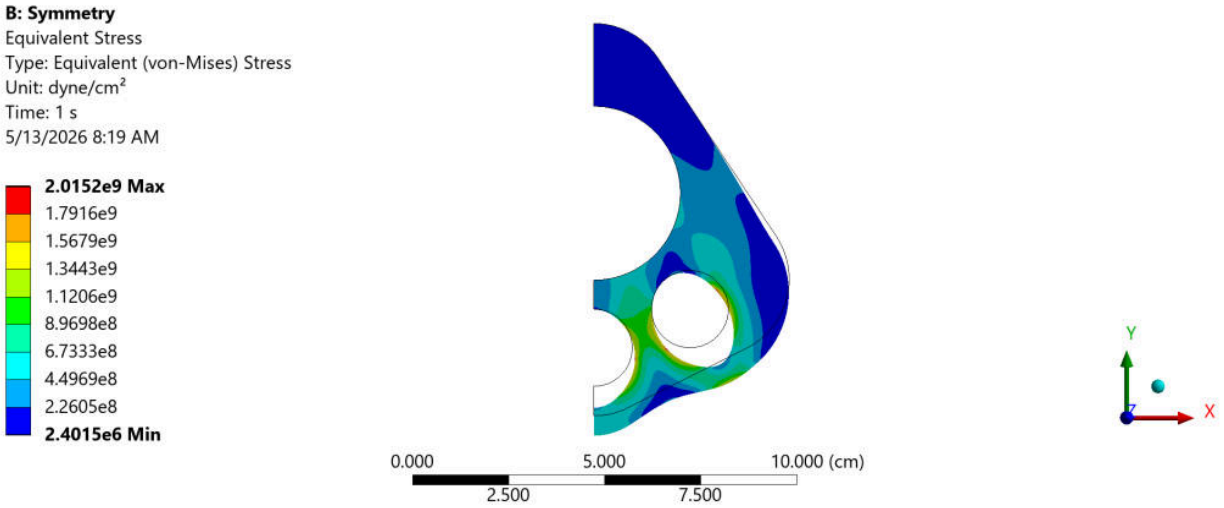
Appendix Figure 3a: 2 cm global stress



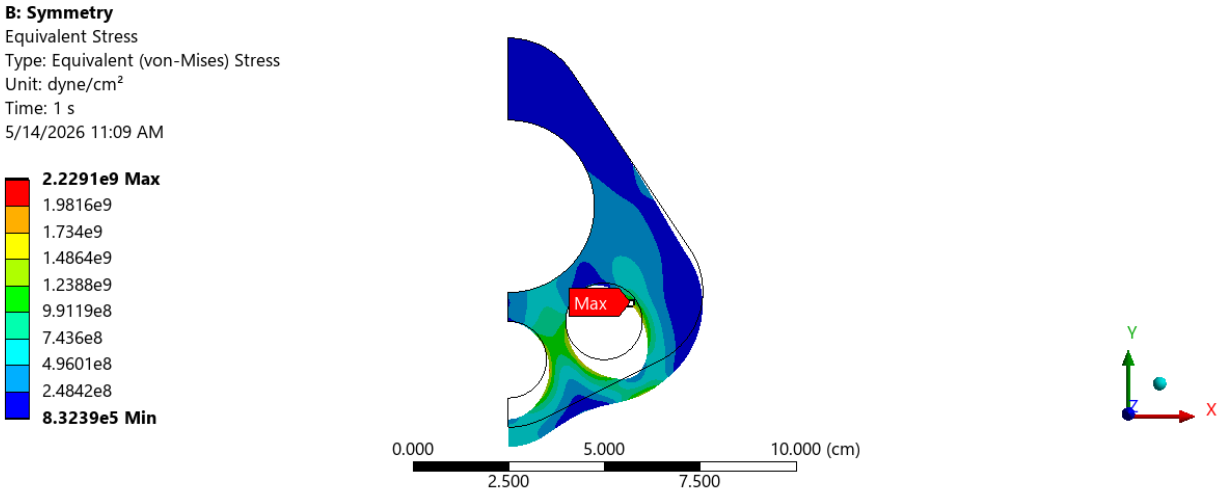
Appendix Figure 3b: 1 cm global stress



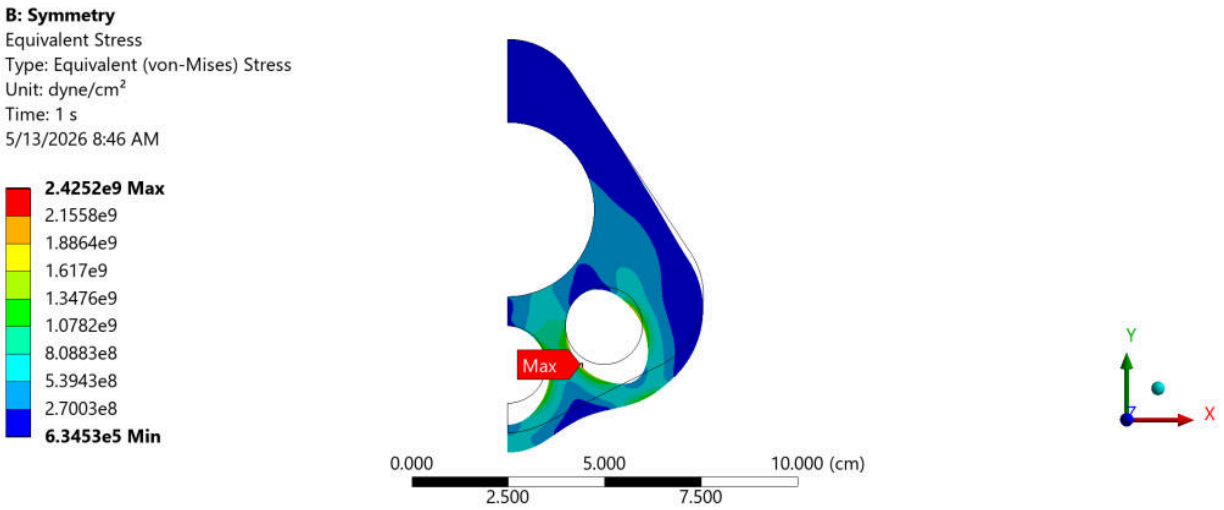
Appendix Figure 3c: 0.2 cm global stress



Appendix Figure 3d: 0.1 cm global stress

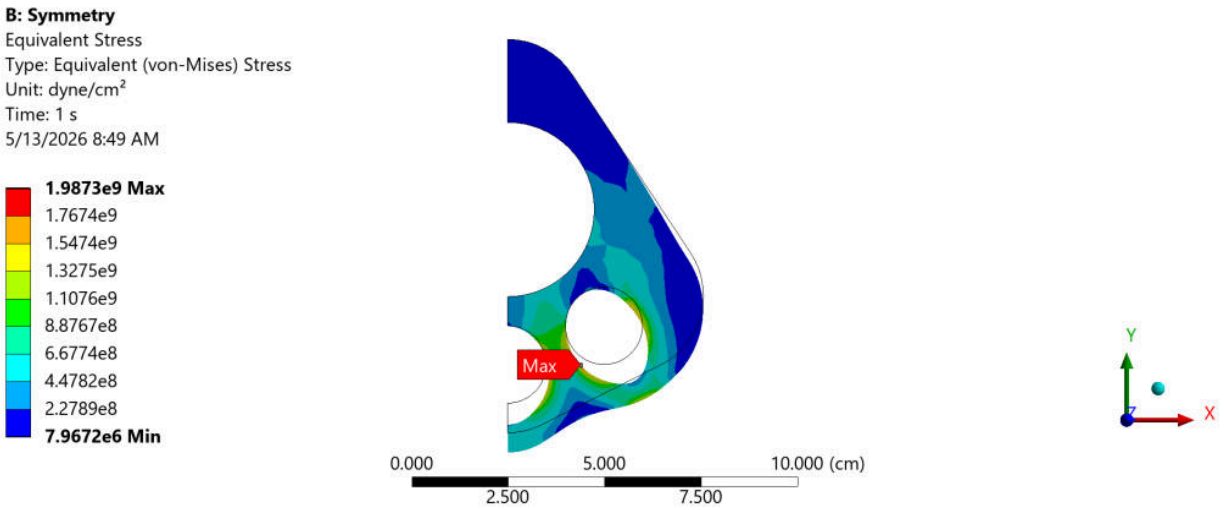


Appendix Figure 3e: 0.04 cm global stress

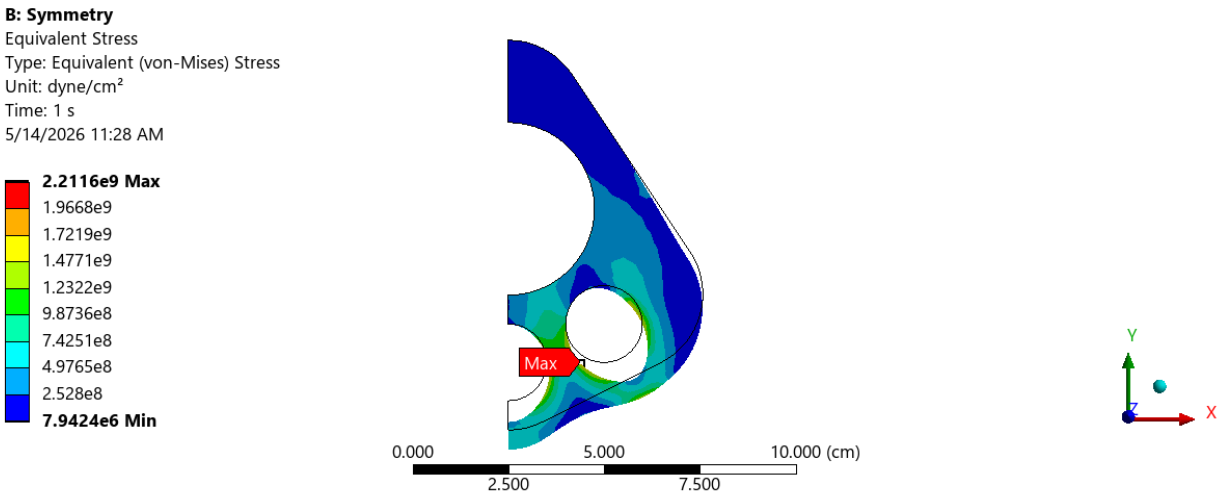


Appendix Figure 3f: 0.02 cm global stress

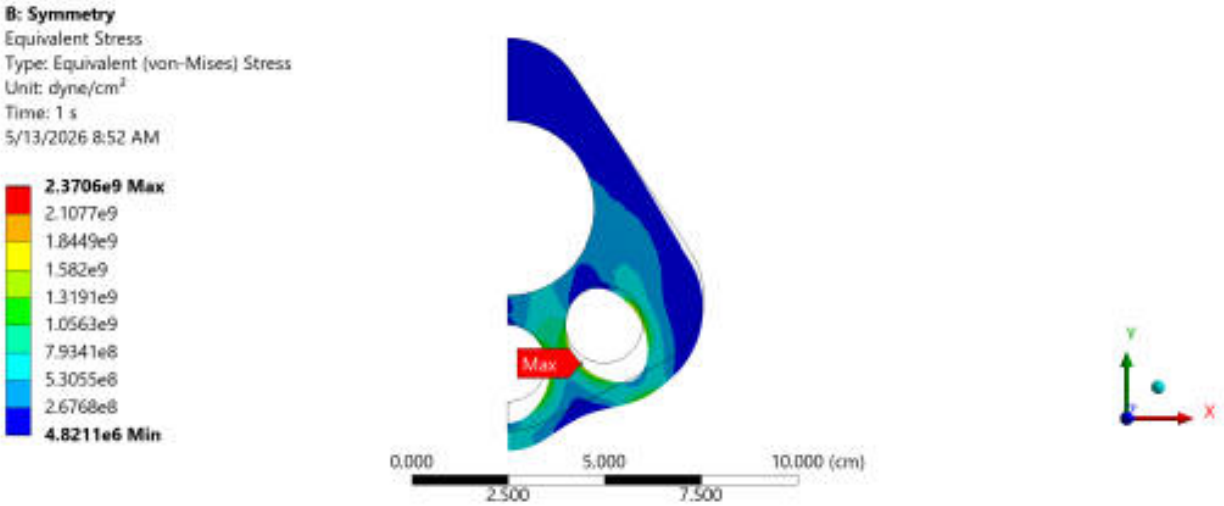
LOCAL STRESSES



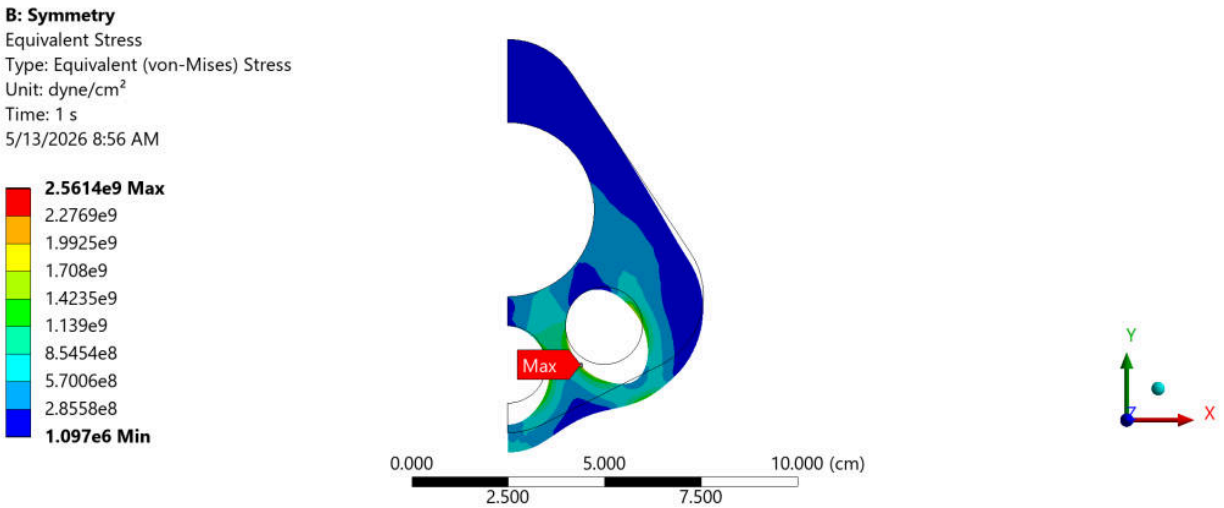
Appendix Figure 3h: 0.1 cm global stress



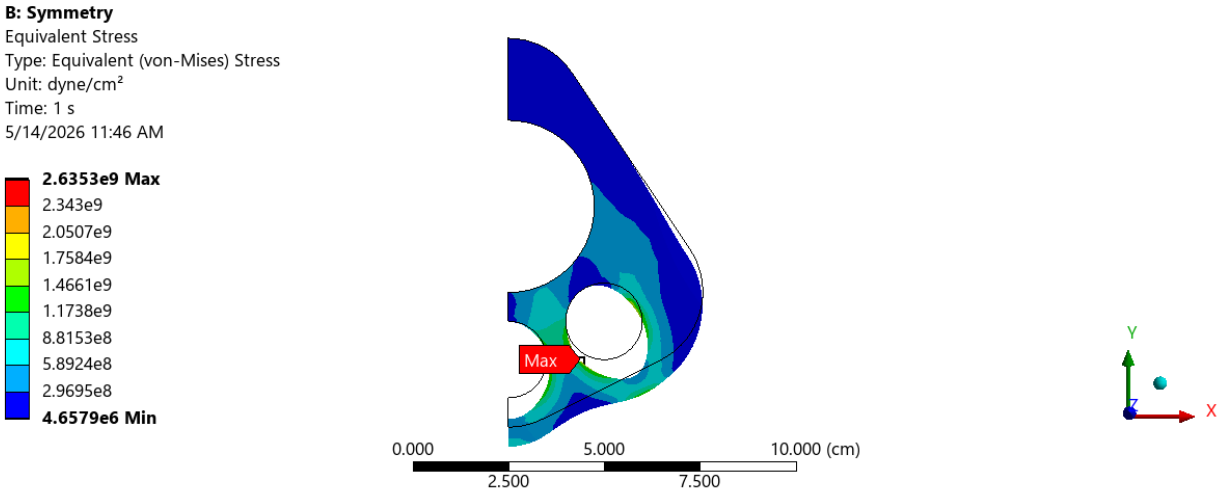
Appendix Figure 3h: 0.04 cm global stress



Appendix Figure 3i: 0.02 cm global stress

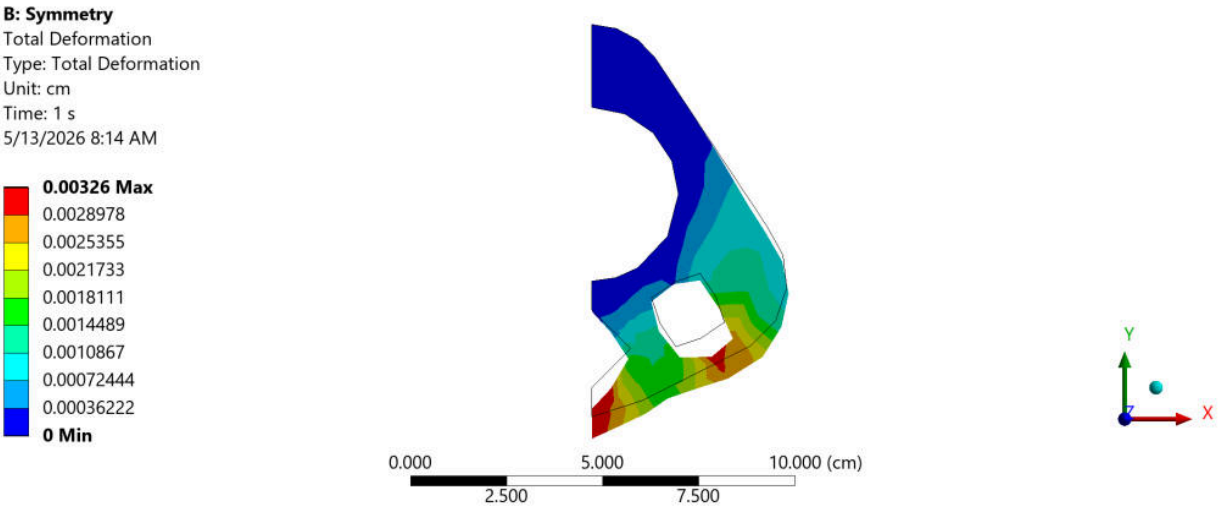


Appendix Figure 3j: 0.01 cm global stress

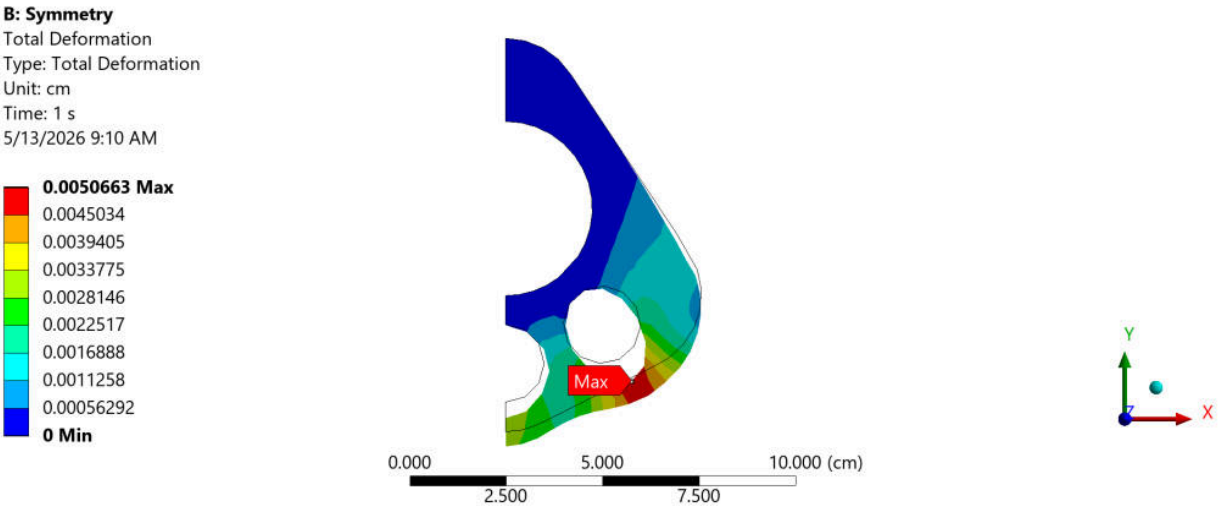


Appendix Figure 3k: 0.008 cm global stress

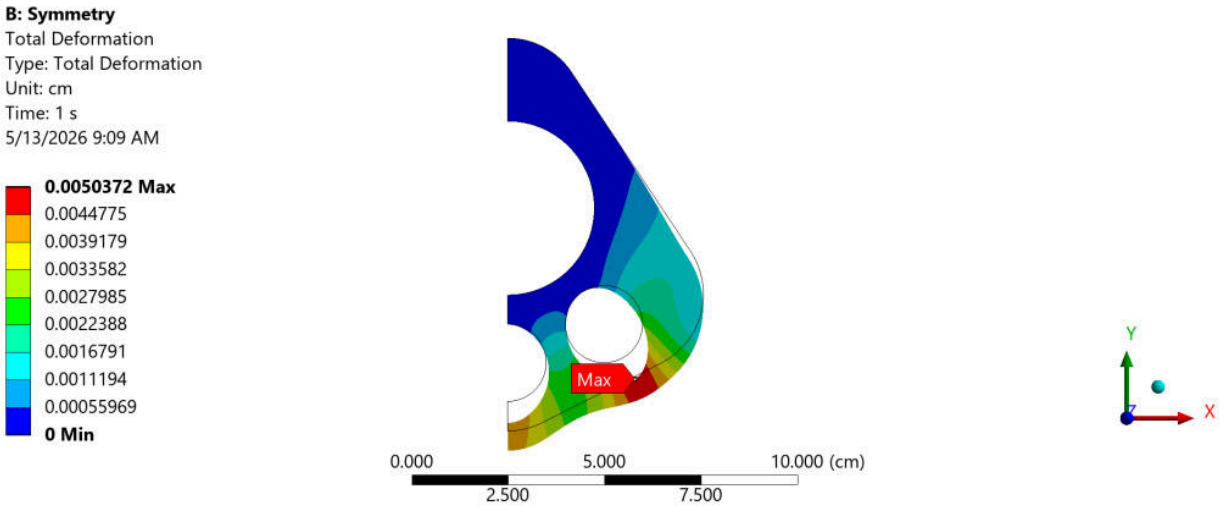
APPENDIX 4: Deformations



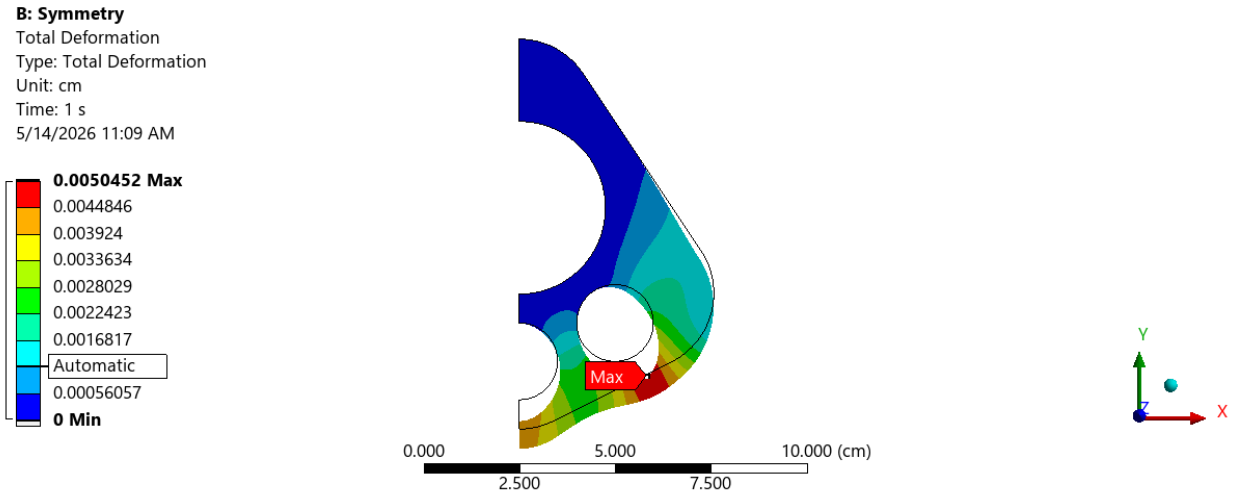
Appendix Figure 4a: 2cm mesh deformation



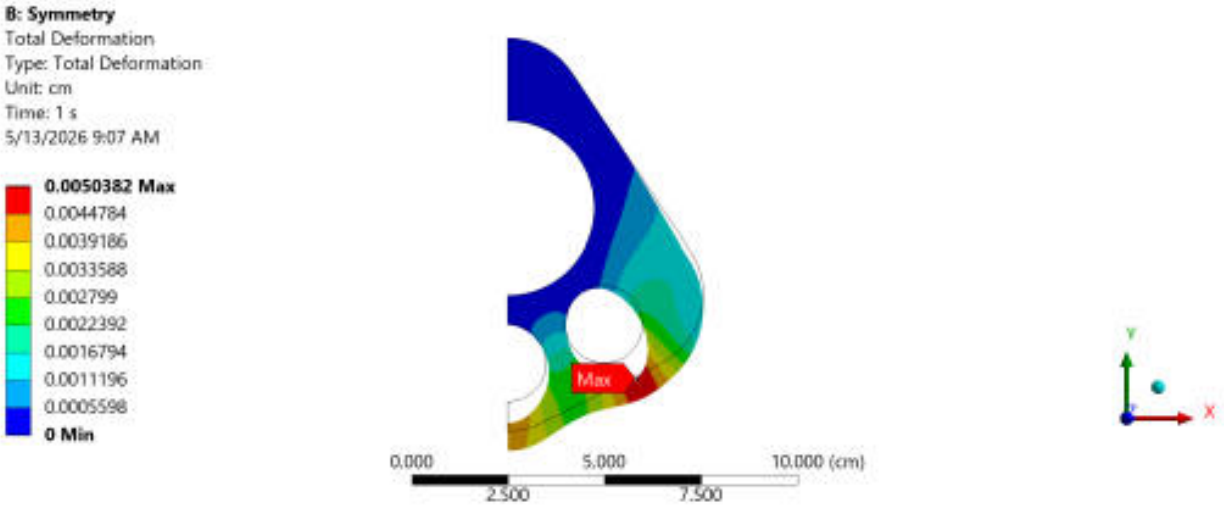
Appendix Figure 4b: 1cm mesh deformation



Appendix Figure 4c: 0.1 cm mesh deformation



Appendix Figure 4d: 0.04 cm mesh deformation



Appendix Figure 4e: 0.01 cm local mesh deformation